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PHYSIOLOGICAL REVIEW

PERIOD3, circadian phenotypes, and sleep homeostasis

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SUMMARY

Circadian rhythmicity and sleep homeostasis contribute to sleep phenotypes and sleep–wake disorders, some of the genetic determinants of which are emerging. Approximately 10% of the population are homozygous for the 5-repeat allele (*PER3*$^{5/5}$) of a variable number tandem repeat polymorphism in the clock gene *PERIOD3* (*PER3*). We review recent data on the effects of this polymorphism on sleep–wake regulation. *PER3*$^{5/5}$ are more likely to show morning preference, whereas homozygosity for the four-repeat allele (*PER3*$^{4/4}$) associates with evening preferences. The association between sleep timing and the circadian rhythms of melatonin and *PER3* RNA in leukocytes is stronger in *PER3*$^{5/5}$ than in *PER3*$^{4/4}$. EEG alpha activity in REM sleep, theta/alpha activity during wakefulness and slow wave activity in NREM sleep are elevated in *PER3*$^{5/5}$. *PER3*$^{5/5}$ show a greater cognitive decline, and a greater reduction in fMRI-assessed brain responses to an executive task, in response to total sleep deprivation. These effects are most pronounced during the late circadian night/early morning hours, i.e., approximately 0–4 h after the crest of the melatonin rhythm. We interpret the effects of the *PER3* polymorphism within the context of a conceptual model in which higher homeostatic sleep pressure in *PER3*$^{5/5}$ through feedback onto the circadian pacemaker modulates the amplitude of diurnal variation in performance. These findings highlight the interrelatedness of circadian rhythmicity and sleep homeostasis.

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Introduction

Sleep is a rich phenotype and individual differences in sleep encompass aspects such as its timing, duration and sleep structure. These differences are observed in the population of healthy individuals and extend into the realm of sleep disorders. For example, inter-individual variation in sleep timing and diurnal preference is considerable within the healthy non-complaining population,[1,2] but in its extremes may lead to a clinically significant complaint, such as advanced or delayed sleep phase disorder (ADSP, DSPD). The mechanisms underlying these individual differences are of great interest, not only because they may provide insight into their functional significance, but also because this may lead to new treatments of the disorders of sleep, including insomnia.[3–5]

The two-process model of sleep regulation has provided a widely accepted conceptual approach to the study of differences in sleep regulation.[6,7] In essence, it states that sleep is regulated though the interaction of two oscillatory processes: the sleep homeostat and the circadian pacemaker. The sleep homeostat is an hourglass oscillator tracking the history of sleep and wakefulness, and thereby tracks sleep debt. Established markers of the sleep homeostat are slow wave activity (SWA) in the EEG during NREM sleep[8] and theta EEG activity during wakefulness.[9] It has been suggested that changes in these markers are related to some of the biochemical consequences of sleep and wakefulness such as variation in extracellular adenosine concentration and other sleep regulatory substances, or related to variation in connectivity, i.e., synaptic strength, in neuronal networks.[10–12] The circadian oscillator, located in the suprachiasmatic nuclei (SCN) of the hypothalamus, is a self-sustained oscillator that determines the preferred timing of sleep and wakefulness.[13] Established markers of the circadian process include plasma melatonin, cortisol and core body temperature.[14]

There is now a wealth of data supporting the essential features of this model.[14–16] Furthermore, the neuroanatomical basis of the circadian regulation of sleep in particular has been elucidated in some detail.[17] In fact, new mathematical models based on this functional neuroanatomy have been developed.[18,19]

We will summarize some of the data in support of the circadian and homeostatic regulation of sleep and waking performance, and discuss how detailed analyses of the interaction of these two processes has provided evidence that, contrary to the predictions of the two-process model, the sleep homeostat feeds back onto the circadian process.[20] We also describe how individual differences in sleep or circadian phenotypes may, theoretically, be related to either of these two processes or their interaction.

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Nomenclature

ASPD	advanced sleep phase disorder
BOLD	blood-oxygen-level dependent
CRY	cryptochrome
DSPD	delayed sleep phase disorder
EEG	electroencephalogram,
fMRI	functional magnetic resonance imaging
mRNA	messenger ribonucleic acid
NREM	non-rapid eye movement sleep
OVLT	organum vasculosum lamina terminalis
PER	period
REM	rapid eye movement sleep
SCN	suprachiasmatic nuclei
SWA	slow wave activity
SWS	slow wave sleep
VLPO	ventral lateral pre-optic area
VNTR	variable number tandem repeat

Genetic factors have been shown to contribute considerably to individual differences in sleep traits such as diurnal preference,[21] or EEG characteristics,[22] but few of the genes that mediate this heritability have been identified. However, the core set of genes that are involved in the generation of circadian rhythmicity have been recognized. The molecular oscillator consists of the positive transcription factors CLOCK and BMAL1, which as a dimer bind to promoter elements of *PERIOD* (*PER*) and *CRYPTOCHROME* (*CRY*) genes and induce their expression. PER and CRY proteins are translated in the cytoplasm, where they can be phosphorylated by Casein Kinase 1, a process that can either target the proteins for F-box-mediated proteosomal degradation, or enhance nuclear translocation, depending upon the site of phosphorylation. PER and CRY proteins form dimers that can translocate to the nucleus, where they provide negative feedback on promotion of their own genes by inhibiting CLOCK/BMAL1-mediated expression. This molecular feedback loop sets the period of the oscillator, which can be governed by post-translational modification, such as phosphorylation.[23,24] Variations in these genes have been related to some individual differences in sleep and circadian phenotypes. Please note that some of the genes involved in the homeostatic regulation of sleep have also been identified. These include genes coding for the adenosine receptors, as well as adenosine deaminase.[25] We will not discuss the contribution of these genes to the homeostatic and circadian regulation of sleep. Here, we will only discuss the impact of variation in one of the clock genes, *PER3*, on sleep and circadian phenomenology, and discuss these effects within the context of the homeostatic and circadian regulation of sleep. One main conclusion derived from these observations and related findings on the effects of other clock genes on sleep homeostasis in animals, is that at the molecular level, circadian rhythmicity and sleep homeostasis are closely interrelated.[26]

Determinants of individual differences in sleep and circadian phenotypes: theoretical considerations

Differences in sleep timing may be related to social factors, such as work schedules, variation in light input as well as variation in the circadian and homeostatic processes (Fig. 1).

It has been established that variation in the timing of sleep is associated with variation in the timing of rhythms driven by the SCN. The core body temperature, cortisol and melatonin rhythm of healthy early sleepers is set to an earlier phase compared to late sleepers. The differences in the timing of these circadian rhythms persist when the sleep–wake cycle, meal cycles and the light–dark cycle are temporarily 'removed' by keeping people awake for 1–2 days under constant routine conditions.[27] Recently, these associations have been confirmed and extended to the rhythm of *PER3* mRNA in leukocytes (Fig. 2).[28] Thus, the timing of endogenous physiological rhythms associates with inter-individual variation in sleep timing in people without sleep complaints.

Analyses of endogenous circadian rhythms in DSPD have provided some evidence that the abnormally late sleep timing in this condition is associated with later than normal rhythms of melatonin and body temperature. The extent of the delay of sleep timing has, however, been reported to be greater than the delay in the physiological rhythms.[29–31] Associations between altered sleep timing and circadian rhythms have also been observed in ASPD.[32]

Although these data indicate an association between inter-individual differences in sleep timing and individual differences in the timing of circadian rhythms, they do not provide insight into the direction of the causality, nor do they provide direct insights into the underlying mechanisms. Theoretically, an earlier timing of

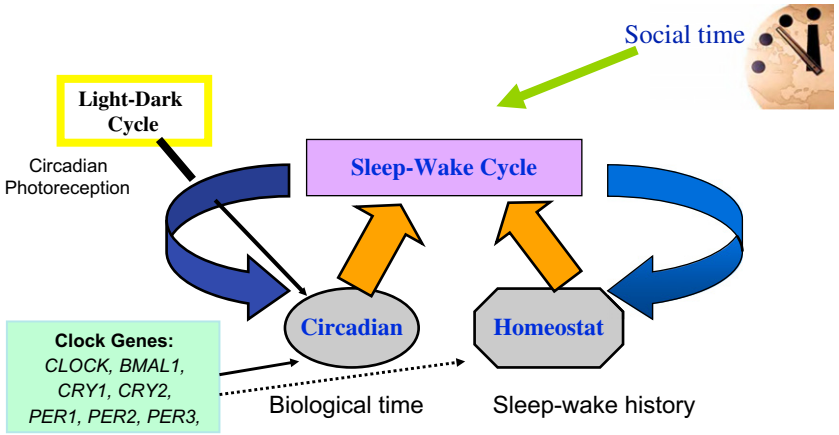


Fig. 1. Elements of sleep–wake regulation. The circadian pacemaker and sleep homeostat are both major determinants of the sleep–wake cycle, which is also affected by social factors such as work schedule. The sleep–wake cycle drives the sleep homeostat and also feeds back onto the circadian pacemaker, which is synchronized by the light–dark cycle. A set of canonical clock genes generate circadian rhythms and some of them have also been shown to affect sleep homeostasis. Modified from.[79]

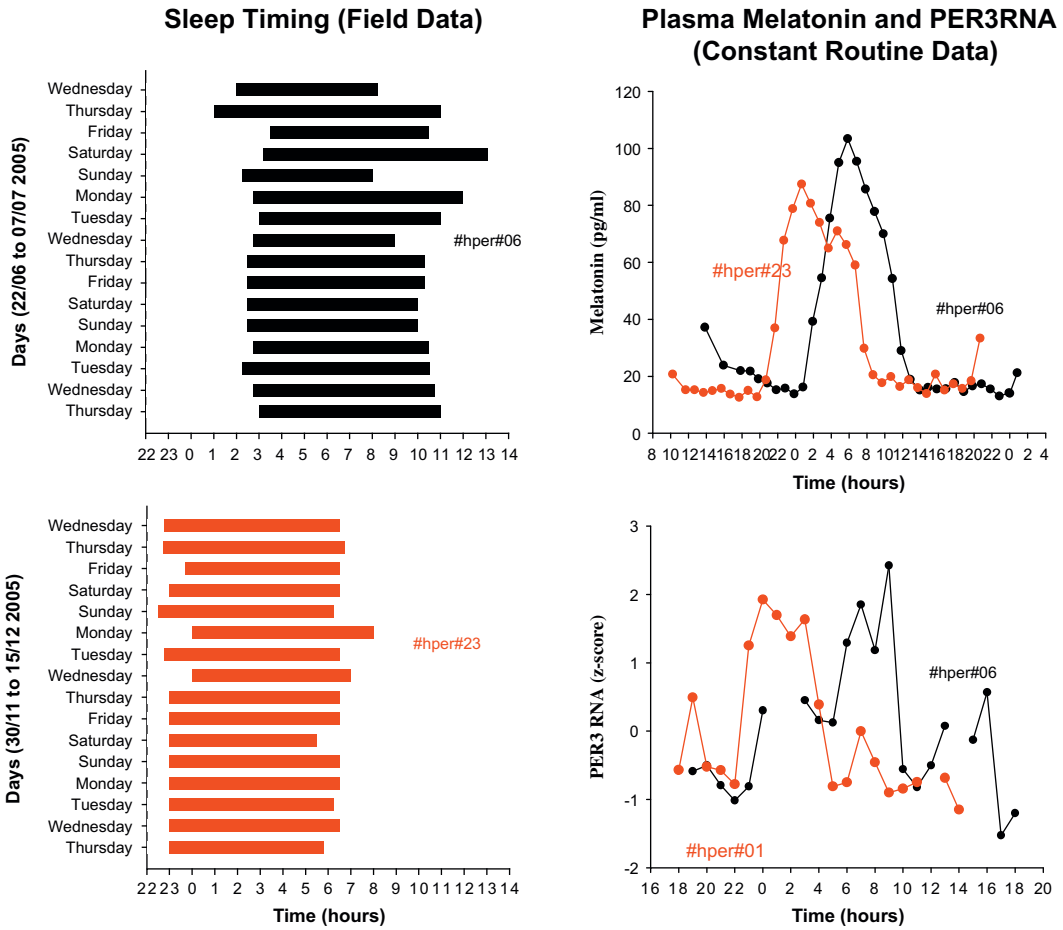


Fig. 2. Habitual sleep timing and rhythms of melatonin and *PER3* RNA. Sleep timing, assessed through actigraphy and sleep diaries, in two subjects while living in their normal environment (Left panels). Plasma melatonin and *PER3* RNA rhythms, as assessed during a constant routine, in early sleepers (#hper#23, #hper#01) and a late sleeper (#hper#06). Based on data published in.[28]

endogenous circadian physiological rhythms could be related to a shorter intrinsic period of the circadian clock, an altered sensitivity of the circadian clock to the synchronizing effects of light, or altered light exposure.

Young adult morning types, who on average have earlier entrained circadian rhythms than evening types, have been shown to have a shorter intrinsic circadian period, as assessed in forced desynchrony protocols.[27] Such an association between morningness and intrinsic period was, however, not observed in older individuals.[33]

A comparison of light sensitivity in morning and evening types is currently not available. Older individuals, who in general are more shifted towards morning preference, are, in general, less sensitive to light than young adults.[34] Light exposure has been shown to be different in morning types compared to evening types[35], and these differences in light exposure, which in part may be mediated by differences in sleep–wake timing, could explain some of the differences in circadian phase. Thus, a variety of circadian factors can contribute to differences in entrained phase and thereby to differences in sleep timing.

Variation in homeostatic processes may also contribute to variations in sleep timing. Theoretically, a reduced build-up of homeostatic sleep pressure will lead to a later onset of sleep, and an accelerated build-up of homeostatic sleep pressure will lead to an earlier onset and offset of sleep. Interestingly, changes in slow wave sleep (SWS) and EEG SWA have been observed in morning and evening types, as well as in DSPD. Morning types display a more rapid increase in theta activity in the waking EEG, indicative of a more rapid build-up of sleep pressure.[36] Furthermore, SWS and SWA in the first NREM period of a sleep episode is enhanced in morning types compared to evening types.[37,38] Mongrain and coworkers have suggested that the morningness/eveningness phenotypes are heterogeneous phenotypes, in which some individuals may be a morning type because of circadian factors, whereas other individuals become a morning type because of homeostatic factors.[39–41]

One prominent example of an identified genetic determinant of abnormal sleep timing is the mutation in the *PER2* gene in an extended family (kindred 2174) with members suffering from ASPD.[42] In this kindred, very early sleep timing appears to be caused by an alteration in the key clock gene *PER2*. So far, the data on sleep and circadian physiology of this very rare mutation is limited. Circadian period has been assessed under classical free-running conditions in one affected 69 year old male of kindred 2174 and was found to be only 23.3 h, which is much shorter than the free-running period in control subjects. Sleep structure, as assessed by visual scoring, was quantified in five individuals affected by ASPD and reported to be not abnormal, even though sleep onset and sleep offset were advanced by as much as approximately 3.75 and 3.5 h, respectively.[43] Based upon the available data, it thus appears that in these families this sleep phenotype is caused by a change in *PER2*, which leads to a change in period and abnormal

entrainment of circadian rhythms, including the sleep–wake cycle, with no effects on sleep structure.

In another kindred, a mutation in the Casein Kinase I delta gene, which also plays a key role in circadian rhythmicity, co-segregates with ASPD. In animal models, this mutation has been shown to affect circadian period but data on effects on sleep structure are not available for either human carriers of this mutation or animals.[44]

PER3 is another member of the family of *PERIOD* genes. It has been identified in mammals,[45] birds,[46] and fish.[47] In mice, it is rhythmically expressed in the SCN, the organum vasculosum lamina terminalis (OVLT), the arcuate nucleus, and the ventromedial hypothalamic nucleus. Expression in the OVLT is strong and in-phase with *Per3* expression in the SCN. This may be of some interest, since the OVLT is involved with hormonal and autonomic regulation. It is also expressed in other brain areas including the gyrus dentatus, medial amygdaloid nucleus, cingulate cortex, hippocampal pyramidal cells, cerebral cortex, and nucleus tractus solitarius.[48] Interestingly, in mice, *Per3* is also expressed in the VLPO, which is a key area in sleep–wake regulation. In fact, expression of *Per3* in the VLPO appears to be in anti-phase with expression in the SCN.[49] Early studies in which the *Per3* gene was disrupted producing no detectable protein, showed significant but minor changes in circadian period.[50] In humans and other primates, the *PER3* gene contains a variable number tandem repeat (VNTR) polymorphism, in which a 54-nucleotide coding-region segment is repeated 4 or 5 times, in the case of humans.[51,52] Several aspects of this polymorphism are of interest. Firstly, it is not present in non-primate mammals.[53] Secondly, the repeated motifs in the protein contain numerous potential phosphorylation sites, which implies that this polymorphism could affect post-translational modification and stability of the protein, in addition to any effect on tertiary structure. Population studies have shown that in most populations around 10% of individuals are homozygous for the 5-repeat ($PER3^{5/5}$), whereas approximately 50% are homozygous for the 4-repeat ($PER3^{4/4}$). In some populations in Papua New Guinea the prevalence of the various genotypes appears to be reversed.[54,55]

PER3: association with diurnal preference

In an association study, the frequency of people homozygous for the 5-repeat was found to be higher in the morning types than in evening types, whereas in DSPD the prevalence of the 5-repeat allele was very low.[52] The association between diurnal preference and the VNTR polymorphism persisted in an extended sample, and there was some evidence that the association was age-dependent [56] The association between *PER3* genotypes and diurnal preference was confirmed in an independent Brazilian study, although the association with DSPD was not confirmed.[57] Because of the relatively high allele frequency, as well as its established association with a classical circadian/sleep phenotype, i.e., diurnal preference, the *PER3* VNTR polymorphism is an interesting model for the study of the impact of clock genes on the homeostatic and circadian regulation of sleep. To establish the impact of this polymorphism requires a prospective approach in which individuals are selected only on the basis of their *PER3* genotype, and not on the basis of a circadian phenotype such as morningness–eveningness.

PER3: sleep timing and mRNA rhythms in leukocytes

In a first approach, we investigated the association between circadian parameters and habitual sleep timing in a group of individuals selected only on the basis of homozygosity for each allele. In both genotypes, we observed robust associations between habitual sleep timing and the rhythms of melatonin, as well as cortisol, as assessed under constant routine conditions. We also investigated associations between habitual sleep timing and the phase of the rhythm of mRNA of *BMAL1*, *PER2* and *PER3* in leukocytes. We found that the association between habitual sleep timing and mRNA rhythms was stronger for *PER3* than the other clock genes, but not as strong as the association with melatonin. When we compared these associations in the two genotypes an interesting difference emerged. The association between circadian markers and sleep/wake timing was more robust in $PER3^{5/5}$ than in $PER3^{4/4}$ (Fig. 3).[28]

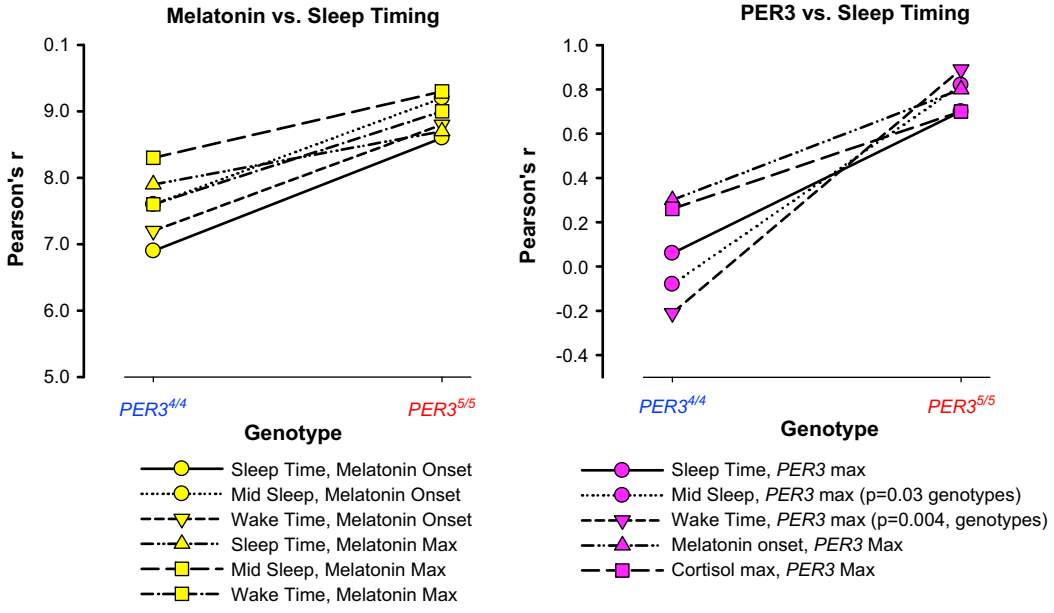


Fig. 3. *PER3* VNTR polymorphism and strength of association between sleep timing and circadian rhythms of melatonin and *PER3* RNA in leukocytes. Correlation between markers of habitual sleep timing (sleep-time, mid-sleep and wake-time) and circadian phase markers (melatonin onset, melatonin maximum, *PER3* RNA maximum, cortisol maximum) as assessed during a constant routine, in $PER3^{5/5}$ and $PER3^{4/4}$. Please note that for all comparisons the correlation is stronger in $PER3^{5/5}$ and significantly so for the correlation between mid-sleep and *PER3*-max and the correlation between wake-time and *PER3*-max. Figure based on Table 4 of.[28]

This finding suggests a more rigid circadian control in *PER3*$^{5/5}$, which is reminiscent of the reported stability and rigidity of sleep–wake timing in morning types.

It should be noted that despite these differences in circadian rigidity, the two genotypes did not differ significantly in circadian phase for any of the markers investigated. We also did not observe differences in circadian amplitude for melatonin, cortisol, nor the mRNA rhythms for *BMAL1*, *PER2* and *PER3*.[28,58]

PER3: sleep and waking EEG

We next characterised aspects of sleep homeostasis by recording sleep and waking performance under baseline conditions and in response to sleep deprivation in a prospective study in which subjects were recruited on the basis of their *PER3* genotype.[58] At baseline, *PER3*$^{5/5}$ individuals displayed many of the sleep characteristics previously observed in morning types. Thus, compared to evening types, they had shorter sleep latency, more SWS, and more SWA, in particular in the first part of the nocturnal sleep episode. Quantitative EEG analyses revealed frequency-specific changes in the EEG during NREM sleep that were very similar to the changes induced by sleep deprivation, and fully consistent with the notion that *PER3*$^{5/5}$ individuals live under a higher sleep pressure (Fig. 4). Further evidence in support of this hypothesis was derived from the analysis of a recovery sleep episode, in which the differences between the two genotypes persisted.

Analysis of the waking EEG during the sleep deprivation period demonstrated a more rapid increase of theta/alpha activity in *PER3*$^{5/5}$, similar to the more rapid increase of theta activity observed in morning types. *PER3*$^{5/5}$ individuals also showed increased slow eye movements during sleep deprivation. In addition, marked differences in the spectral power density in the alpha range during REM sleep were observed between the genotypes. *PER3*$^{5/5}$ individuals have much higher alpha activity in REM sleep and wakefulness than *PER*$^{4/4}$. It is unclear how this striking genotype-dependent difference in the REM sleep EEG relates to homeostasis. It has been reported that REM sleep deprivation,[59] as well as repeated partial sleep deprivation,[60] leads to a reduction in alpha activity in REM sleep. Alpha activity could, therefore, be considered a marker of REM sleep homeostasis. However, alpha activity in REM sleep is increased in *PER3*$^{5/5}$, suggesting that REM sleep pressure is lower, rather than higher, in this genotype. Whatever the correct interpretation of the change in the REM sleep EEG may be, it is of interest that the EEG of the genotypes also differed in this frequency range during wakefulness and it has long been known that alpha activity during wakefulness is highly

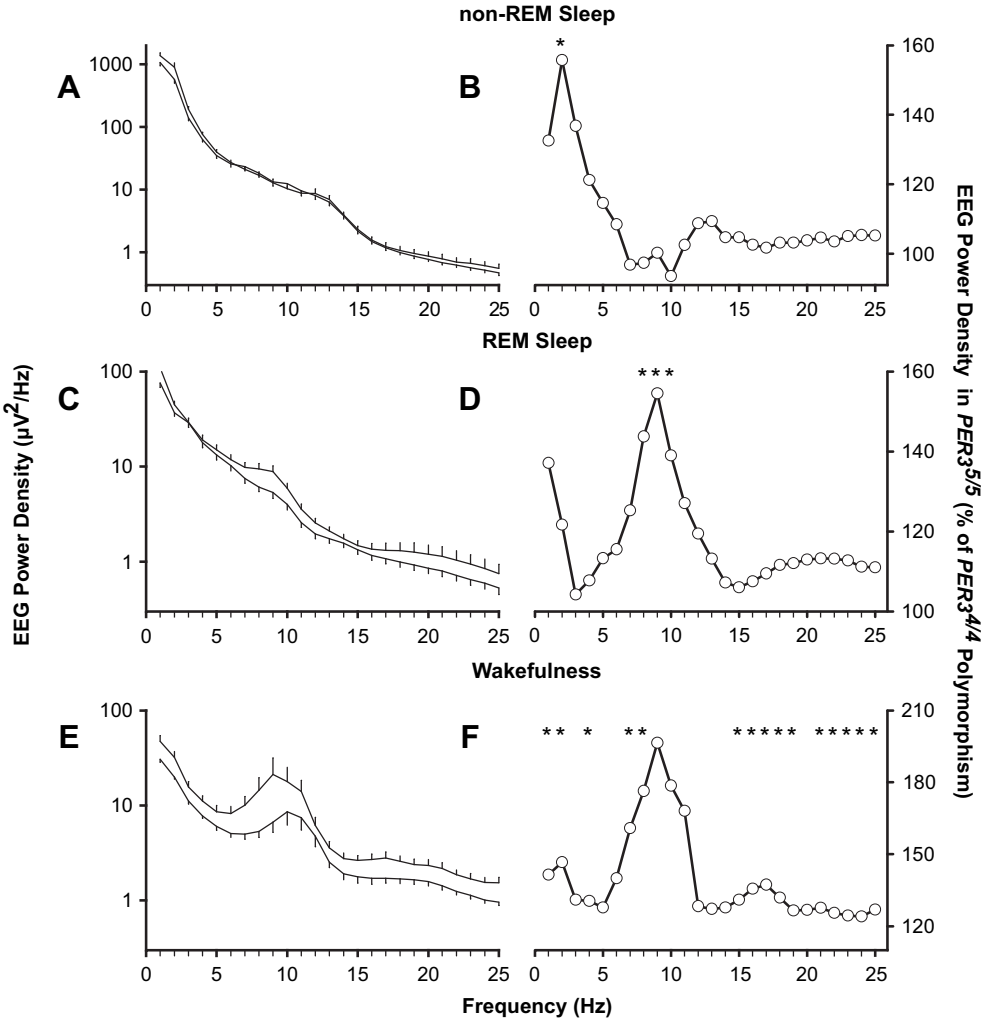


Fig. 4. *PER3* VNTR polymorphism and EEG activity in wakefulness, REM sleep and NREM sleep. Absolute power spectra (left) and relative power spectra (right) in *PER3*$^{5/5}$ and *PER3*$^{4/4}$ individuals. For the relative power spectra, 100% = *PER3*$^{4/4}$. Please note the marked increase in alpha activity in *PER3*$^{5/5}$ in both Wakefulness and REM sleep, as well as the increase in delta activity in NREM sleep. From[58], with permission.

heritable. The *PER3* polymorphism had other effects on sleep and circadian physiology as well. The suppression in REM sleep, which is normally observed after sleep deprivation in young subjects, was more pronounced in $PER3^{5/5}$ individuals.[58] This genotype-dependent difference in REM sleep regulation can be interpreted within the context of the homeostatic regulation of slow wave sleep: more SWS leads to a suppression of REM sleep.[6] Whether all changes in REM sleep associated with this polymorphism are reconcilable with a simple homeostatic interpretation, or whether other explanations, such as those related to the circadian regulation of REM sleep,[14] need to be invoked, will require further perturbations of the homeostatic and circadian processes in these genotypes.

PER3: autonomic regulation of the heart

The autonomic regulation of the heart is modulated by vigilance state and circadian phase. In particular, in the course of a NREM/REM sleep cycle, the sympathovagal balance changes dramatically, such that sympathetic dominance is lowest during NREM sleep and highest during REM sleep. Comparing the time course of the sympathovagal balance during baseline sleep and during recovery sleep after sleep deprivation revealed differences between $PER3^{5/5}$ and $PER3^{4/4}$ individuals, such that the amplitude of the ultradian modulation of sympathovagal balance was lower in $PER3^{5/5}$.[61] These changes in autonomic control of the heart are reminiscent of changes induced by sleep restriction.[62]

PER3: effects on cognitive performance

We next quantified the time course of waking performance during sleep deprivation in the two genotypes. Because it was initially our desire to characterise overall waking performance, we computed a composite score based on verbal and spatial 1-, 2-, and 3-back tests; a sustained-attention-to-response task; a paced-visual-serial-addition task; a self-paced digit-symbol-substitution test; simple-reaction-time and serial-reaction-time tests; and a motor-tracking task, thereby covering a wide-range of cognitive functions. Whereas during the baseline day no major differences between the two genotypes emerged, during sleep deprivation, and in particular during the morning hours approximately 2–6 h after the melatonin peak, performance deteriorated markedly in the $PER3^{5/5}$ individuals, whereas in the $PER3^{4/4}$ individuals this decline was more modest. Following this early morning deterioration in performance, the difference between the genotypes became smaller (Fig. 5).

Thus, the apparent circadian amplitude of waking performance under constant routine conditions is much greater in $PER3^{5/5}$ than in $PER3^{4/4}$.[58] Subsequent analyses of the individual performance tasks revealed that those tasks probing executive function were most affected by the sleep deprivation in the vulnerable genotype. Thus, more demanding versions of both the spatial and verbal N-back tasks and the paced-visual-serial-addition task were performed consistently worse by the $PER3^{5/5}$ individuals during the circadian trough.[63] This is of some interest because executive functions have been considered to be most sensitive to the effects of sleep deprivation in general.

PER3: fMRI-assessed brain responses

All of the above findings, which were based on data collected in one group of subjects, are consistent with the notion that the *PER3* VNTR affects the homeostatic regulation of sleep and that this difference in the homeostatic regulation of sleep, in interaction with the circadian rhythmicity, underlies the differential susceptibility to the negative effects of sleep deprivation on performance.

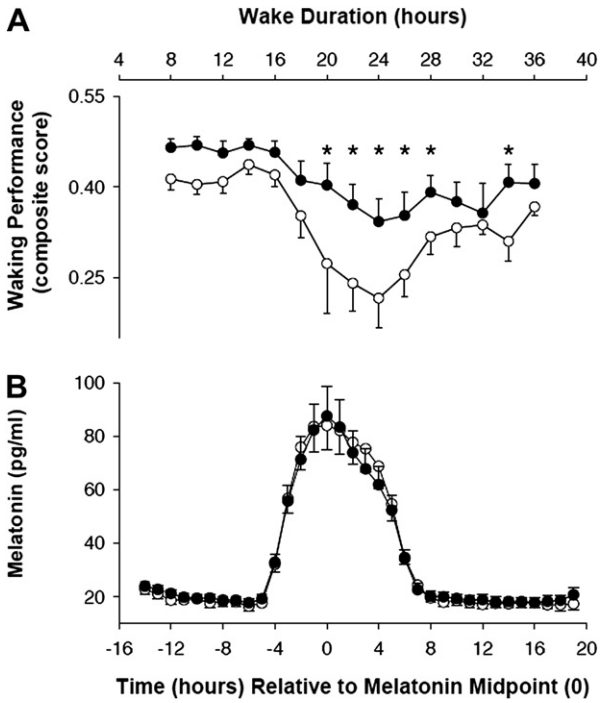


Fig. 5. *PER3* VNTR polymorphism and the decline of cognitive performance during sleep deprivation and circadian misalignment. A composite performance measure, based on verbal and spatial 1-, 2- and 3-backs, sustained-attention-to-response task, paced visual-serial-addition task, digit-symbol-substitution test, simple-reaction-time, serial-reaction-time and a motor-tracking task, declines during sleep loss, but much more so in $PER3^{5/5}$ than in $PER3^{4/4}$, and in particular during the early morning (Panel A). These differences were observed in the absence of a difference in melatonin phase (Panel B). ($PER3^{5/5}$; Open symbols), $PER3^{4/4}$ Closed symbols. Modified from[58], with permission.

To further substantiate this hypothesis, we proceeded in two ways. We first investigated whether the differential response to sleep loss could be replicated in an independent cohort of $PER3^{5/5}$ and $PER3^{4/4}$ individuals, in another prospective study. In this study, our dependent variables were derived from fMRI-assessed brain responses to an executive task in the evening and morning during a normal sleep–wake cycle, and in the morning after a night without sleep. In this experiment, we observed that in the course of a normal waking day, the brain responses to an executive task declined in $PER3^{5/5}$ individuals in a frontal area implicated in executive control. By contrast, in $PER3^{4/4}$ individuals this decline was not observed. Furthermore, whereas in the morning after sleep no major differences in fMRI-assessed brain responses between the genotypes were observed, in the morning after sleep loss major genotype-dependent differences in brain activity were observed. In $PER3^{4/4}$ individuals, sleep loss was associated with an increase in BOLD responses in many frontal and parietal areas. By contrast, in $PER3^{5/5}$ individuals sleep deprivation led to a reduction in BOLD responses in many frontal and parietal areas.[64] Thus, these fMRI data confirm the differential response to sleep loss in the two genotypes consistent with our previous observations. These data also lend further support to the notion that the *PER3* VNTR is a genuine genetic marker for individual differences in the vulnerability to sleep loss, and that this differential vulnerability is related to the interaction of sleep homeostasis and circadian rhythmicity.

PER3: modelling the genotype-specific interaction between the circadian and homeostatic processes

To further strengthen the conceptual framework in which to interpret these findings, we analysed and interpreted our data within the context of our knowledge about the interaction of circadian and homeostatic processes. We first established that the observed differences in SWS and SWA reflect a difference in parameters of the homeostatic process, and are not secondary to the small and statistically not significant differences in sleep duration between the two genotypes. In our original study, $PER3^{5/5}$ individuals slept on average 43 min less than $PER3^{4/4}$ individuals.[58] According to the standard parameters of two-process model of sleep regulation,[65] this should lead to a level of SWA at the onset of baseline sleep of 104% (100% = SWA at the onset of baseline sleep in $PER3^{4/4}$), which is much smaller than what was observed when fitting the data. We have illustrated the estimated time course of the homeostatic process in Fig. 6.

The observed genotype-dependent differences in SWA were much greater than those predicted by the differences in sleep duration. Hence, the most parsimonious explanation for the difference in SWS and SWA are different time constants for the build-up and dissipation of sleep pressure in the two genotypes. To investigate the consequences of the differences for the time course of performance during a baseline day and during sleep deprivation, we developed a simple conceptual model for the regulation of performance by homeostatic and circadian processes. Although this approach is closely related to the two-process model of sleep regulation and subsequent models,[66] it differs in a few essential aspects. Most importantly, it is assumed that the homeostatic process feeds back onto an output immediately downstream from the circadian process (or directly to the circadian process itself). Thus, in contrast to the assumptions of the two-process model of sleep regulation, we assume that the two processes interact. Such an interaction is consistent with the observed dependency of the circadian amplitude of performance on homeostatic sleep pressure, which was first reported in 1992[67] and subsequently confirmed in many studies.[16] We further assume that the circadian process itself consists of wake-promoting and sleep-promoting signals, in accordance with observations from forced desynchrony protocols[68] and analyses of the effects of SCN lesions.[69] The wake-promoting signal reaches its maximum in the wake-maintenance zone and dissipates rapidly afterwards. The sleep-promoting signal reaches its maximum at approximately 2–4 h after the melatonin peak, at approximately 06:00–08:00 h. An increase in homeostatic sleep pressure leads to an inhibition of the wake-promoting signal. Evidence for this has been observed in animal studies[70,71] and in a comparison of fMRI responses during a psychomotor vigilance task in morning and evening types.[72] We further assume that increased homeostatic sleep pressure leads to an enhancement of the sleep-promoting signal. Homeostatic sleep pressure will also affect cortical networks, and performance on executive tasks in particular will ultimately be determined by the interaction of the circadian output and cortical networks. Within this conceptual framework, the time course of variables, such as performance, that result from a multiplicative interaction of circadian and homeostatic processes, can be different in the course of a normal waking day and during sleep deprivation, despite identical core circadian oscillators, provided that the homeostatic process differs between the two genotypes. A cartoon of the circadian and homeostatic processes in $PER3^{5/5}$ and $PER3^{4/4}$ individuals, as well as a time course of performance in these two genotypes is illustrated in Fig. 7.

It shows that the homeostatic oscillator has a greater amplitude in $PER3^{5/5}$ individuals, but that the phase and amplitude of the core circadian oscillator does not differ between the two genotypes. Because the homeostatic oscillator feeds back onto the output of the circadian oscillator, this output differs between the two genotypes. The time course of performance during a normal waking day shows only minor differences between the two genotypes, such that performance declines in $PER3^{5/5}$ and improves in $PER3^{4/4}$. Through the interaction of sleep homeostasis and circadian rhythmicity, sleep deprivation markedly enhances the differences between the two genotypes, particularly in the morning hours. Thus, in accordance with the data, the greatest differences between the genotypes are observed in the morning hours after sleep deprivation and become smaller thereafter. It should be noted that within this model we assume that all relevant homeostatic processes for cognitive performance are linked to the time course of SWA, and not to homeostatic processes related to sleep duration. Repeated partial sleep deprivation studies have suggested that the time course of the effects of sleep restriction on performance dissociates from the effects on SWA.[73,74] Experiments in which sleep is extended or restricted prior to an extended period of wakefulness covering an entire circadian cycle, will have to be conducted in *PER3* genotypes to further investigate the interrelations between sleep duration, sleep homeostasis, SWA and the circadian amplitude of cognitive

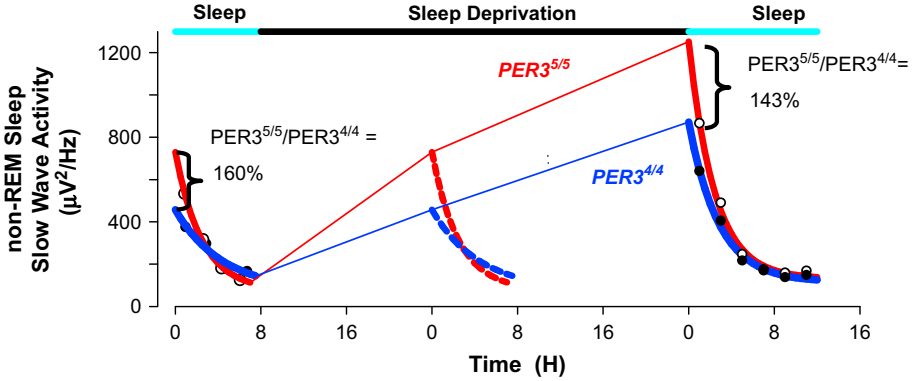


Fig. 6. *PER3* VNTR polymorphism and the time course of homeostatic sleep pressure. The decline of homeostatic sleep pressure during baseline sleep (left) and recovery sleep (right) was estimated by fitting SWA data of frontal EEGs to an exponential function $H_t = H_0 * e^{-T/Tcs} + H_{00}$: where $H_0 = H$ at sleep onset and H_{00} = lower asymptote. T is time since sleep onset. The build-up of homeostatic sleep pressure is indicated by connecting the estimated values at the end of baseline sleep, the beginning of baseline sleep and the beginning of recovery sleep, separately for $PER3^{5/5}$ and $PER3^{4/4}$. Please note the more rapid build-up of homeostatic sleep pressure in $PER3^{5/5}$. Compared to $PER3^{4/4}$ (=100%), homeostatic sleep pressure in $PER3^{5/5}$ is higher during both baseline sleep (160%) and recovery sleep (143%). Modified from[26], with permission.

decline. This should, however, not affect the core assumption of our model, which is that a change in sleep homeostasis may lead to a circadian phenotype. Within this conceptual framework, the effects of the *PER3* VNTR polymorphism on sleep and circadian phenotypes appear to be mediated through effects on sleep homeostasis. This may be just one example of a larger set of phenomena that could be explained in a similar way. For example, animal studies have documented that alterations in other clock genes such as *Per2*, *Cry1*, *Cry2* and *Npas2*, alter homeostatic aspects of sleep regulation, even though they do not completely abolish the homeostatic response to sleep loss.[26]

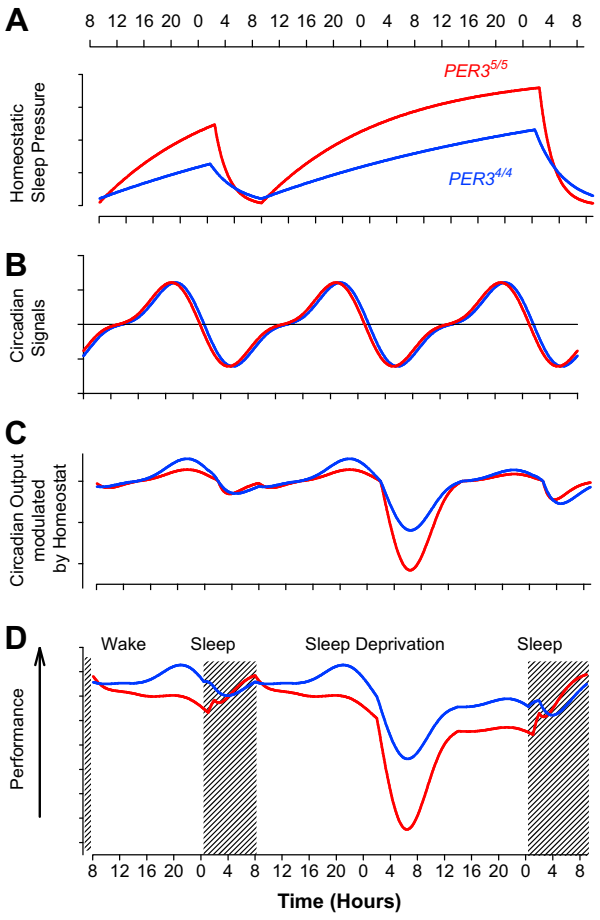


Fig. 7. *PER3* VNTR and the interaction of the circadian and homeostatic processes in the regulation of performance. Conceptual model for the regulation of sleep–wake and performance in $PER3^{5/5}$ (red) and $PER3^{4/4}$ *(blue)*. **Panel A**: The homeostatic process H increases during wakefulness and declines during sleep. The time constants of this process are shorter in $PER3^{5/5}$ than in $PER3^{4/4}$ and, therefore, the amplitude of the H oscillation during a normal sleep–wake cycle (left side of panel) is greater in $PER3^{5/5}$. During sleep deprivation (right side of panel), there is a prolonged increase in homeostatic sleep pressure, followed by its return to baseline during recovery sleep. **Panel B**: A circadian signal promoting wakefulness and sleep does not differ in either phase or amplitude, between the genotypes. **Panel C**: A circadian output signal modulated through feedback of H on C. Please note that at the end of the waking day, the attenuation of the wake-promoting signal by homeostatic sleep pressure is greater in $PER3^{5/5}$ than in $PER3^{4/4}$; during sleep deprivation, the sleep-promoting signal, which is maximal in the morning hours, is amplified by homeostatic sleep pressure and more so in $PER3^{5/5}$ than in $PER3^{4/4}$. **Panel D**: Performance, which is a simple function of the circadian output signal (C modulated by H) and H, is near stable during a normal waking day, although a small decline is observed in $PER3^{5/5}$ (typical for morning types) and an increase in $PER3^{4/4}$ (typical for evening types). During sleep deprivation, performance is poorest in the early morning hours and particularly so in $PER3^{5/5}$. Modified from[80], with permission.

Practice points

- Differences in sleep timing may be related to either homeostatic or circadian processes.
- Differences in the susceptibility to the effects of sleep loss may also be related to either homeostatic or circadian processes.
- The *PER3* VNTR is a genetic marker to the susceptibility to the effects of total sleep deprivation and circadian misalignment.
- Homozygosity for the 5-repeat is associated with more rapid build-up of sleep pressure during sleep loss.
- A conceptual model shows that this difference in sleep homeostasis through its feedback on circadian processes can lead to observed differences in performance.

Research agenda

Understanding sleep and circadian phenotypes requires elaborate quantification of the rich variety of sleep and circadian processes and markers thereof. Often sleep and circadian phenotypes are typecast as belonging to one or the other without careful description. In terms of future research, better characterisation of sleep and circadian phenotypes also with respect to their response to sleep and circadian perturbations is recommended. This may require protocols of sleep restriction and sleep extension, as well as circadian misalignment.

It will be important to determine to what extent the effects of the *PER3* VNTR on performance are restricted to a particular circadian phase and to what extent the effects on the sleep EEG are dependent on the type of sleep loss, i.e., total sleep deprivation or repeated partial sleep loss. For example, one recent post-hoc analysis of the response to four nights of sleep restriction (4 h) in *PER3* VNTR genotypes confirmed an effect of the *PER3* polymorphism on the homeostatic response as indexed by Slow Wave Energy. However, an effect of the *PER3* polymorphism on cognitive decline during the daytime in response to this sleep restriction was not observed in this study.[75]

Furthermore, the description of the effects of variation in known clock and sleep genes should, if at all possible, include a wide-range of sleep, circadian and performance variables. For example, in addition to the phenotypes that we describe here, the *PER3* VNTR polymorphism has also been associated with risk of developing breast cancer,[76] age of onset in mood disorders,[77] and heroin dependence.[78]

Understanding the mechanisms by which polymorphisms such as the *PER3* VNTR bring about their effects will require characterisation of the molecular and cellular consequences of these genetic variations. Understanding these consequences at that level of description may also provide insight into the basis and function of the homeostatic regulation of sleep. A better understanding of the processes underlying the effects of genetic variation on sleep and circadian phenotypes may lead to the development of new countermeasures and therapies.

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